

Experiments with a Spectrometer

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Abstract

The wavelengths of light emitted by a hydrogen lamp were measured using a diffraction grating with slit spacing of 300 lines/mm by measuring the angle of diffraction of the emitted light. The Rydberg constant was calculated using the wavelengths of the first three emission lines of the Balmer series for Hydrogen. The Rydberg constant was found to be $(1.209 \pm 0.5) \times 10^7$. This is consistent with the accepted value of 1.097×10^7 . The refractive index of a glass prism was measured for different wavelengths of light by measuring the angle of incidence and refraction of the light. The refractive index of the prism and wavelength of the light was then used to validate Cauchy's relationship, which was found to be consistent with the results obtained.

1. Introduction

A spectrometer is an instrument used to measure properties of electromagnetic radiation. The instrument used in this experiment measures the wavelengths of atomic lines in the optical region. The Rydberg constant represents the inverse wavelength of the lowest energy photon capable of ionising the hydrogen atom from its ground state. The Rydberg constant was named after J.R. Rydberg in 1890 when he used it as a fitting parameter for the Balmer formula to include all spectral series of lines for the hydrogen atom. Niels Bohr later showed that its value can be calculated from other physical constants based on his atomic model, see ref [1].

2. The spectrometer

The spectrometer in this experiment consists of a slit that is at the focal point of a condensing lens in the collimator. A hydrogen lamp is positioned so the slit is illuminated which produces a parallel beam of light. This beam of light is dispersed by a diffraction grating or prism positioned on a rotating table mounted on the base of the spectrometer. The objective lens of the telescope brings the parallel beam of light into focus in the plane of the cross hairs. The telescope rotates about the same axis as the prism. The angle of rotation is measured using two vernier scales.

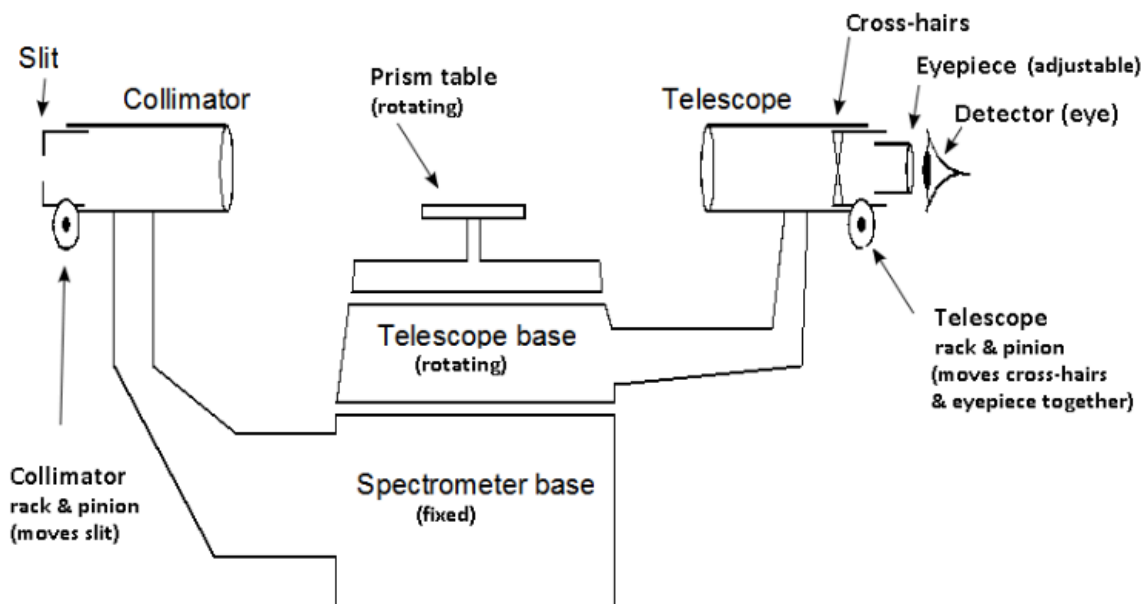


Fig 1. The spectrometer. Both the prism and telescope tables have a locking mechanism and fine adjustment

3. Diffraction grating and Rydberg constant

The spacing of a diffraction grating was used to measure the wavelengths of lines of hydrogen in the Balmer series. The wavelengths are given by the Rydberg formula,

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right), \quad (1)$$

where n_1 and n_2 are the principle quantum numbers of the atomic states, see ref [2]

The light incident at an angle i on the diffraction grating was diffracted through an angle r . The wavelength of the light is related to these angles by,

$$\sin i + \sin r = \frac{m\lambda}{a} \quad (2)$$

where a is the line spacing and m is the diffraction order.

3.1 Initial setup

The spectrometer was setup as shown in fig 1. The eyepiece was adjusted so that the cross hairs were in focus about 45 degrees to the image of the slit. The telescope was adjusted so that both the hydrogen lamp and the cross hairs were in focus at the same time, setting the telescope for parallel light.. The collimator was adjusted so that the image of the slit was in focus, setting the collimator to produce a parallel beam of light originating at the slit. The condensing lens was positioned so the light from the hydrogen lamp fell on the slit. The lens, slit and collimator were adjusted to be the same height, ensuring that the hydrogen lamp was on the optic axis of the collimator.

3.2 Diffraction grating

The diffraction grating used in this experiment was a grating with 300 lines per millimetre, giving a slit spacing of 3.3×10^{-6} metres. The grating was mounted on the rotating table of the spectrometer, with the lines of the grating parallel to the axis of rotation. The grating was set so that the angle of incidence to the parallel beam was zero. This was done by recording the angle of the telescope when set to view the slit directly. This angle was recorded and called the undeviated angle, it was equal to 56.483 ± 0.004 degrees. The telescope was then rotated 90 degrees and clamped. The grating was inserted and rotated until the reflected image of the slit was observed in the telescope. The grating was then rotated 45 degrees and clamped.

Different coloured lines were observed as the telescope was moved around the axis of rotation. The different colours correspond to the different atomic transitions in the Balmer series.. The angle of diffraction r for each line was recorded by positioning the cross hairs of the telescope in the centre of each line. The angle of the telescope was recorded for both lines to the left and right of the undeviated angle and the average was taken. The angles of diffraction for each of the three emission lines -purple, green and red in this experiment- were taken for each of three orders of

diffraction 1, 2 and 3. Values of r and subsequently $\sin(r)$ were found for each emission line at each order, where r was found by subtracting the angle of the undeviated telescope from the average angle of diffraction.

The angle was read off two Vernier scales 180 degrees apart and the average of the two taken to give a result (after subtracting 180 degrees from one). This allows for correction of the offset between the axis of rotation and the position of the scale. It reduces the chance of a systematic error.

3.3 Determining the Rydberg constant

The values for the average angles of diffraction r for the diffraction grating are shown in Fig. (3). The uncertainty on the recorded angles was taken to be ± 0.004 as this was the resolution of the Vernier scale. The uncertainty for the average was then calculated to be ± 0.006 by combining in quadrature. These results show that shorter wavelengths of light have smaller angles of diffraction.

	Angle of diffraction (degrees)		
	First order	Second order	Third order
Purple	63.450	71.037	74.410
Green	64.700	73.333	82.355
Red	67.854	79.342	92.615

Fig 3. Table showing angles of diffraction

The graphs of $\sin(r)$ plotted against order can be seen in Fig.(4). The uncertainty on $\sin(r)$ was calculated using the propagation of error, by using this formula the uncertainty was found to be $\cos(r)$ times the uncertainty on r , giving 0.004 for each gradient. From these graphs the respective wavelengths for the purple, blue and red emission lines were calculated as the gradient of the graph is $\frac{\lambda}{a}$.

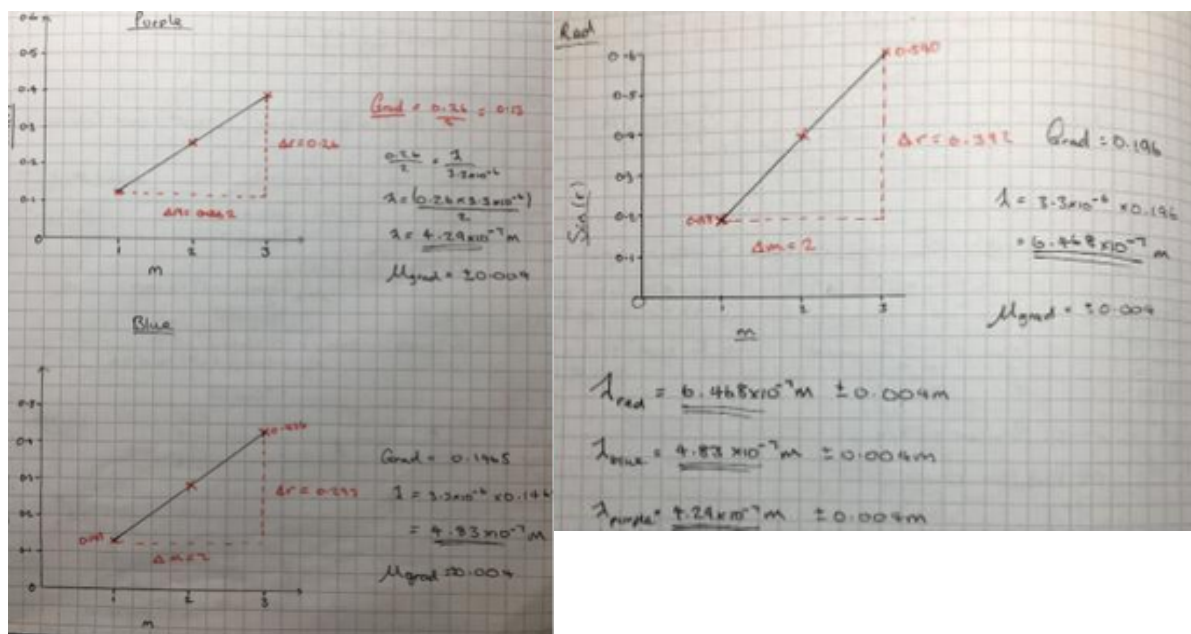


Fig 4. Graphs of $\sin(r)$ against order of diffraction

A plot of the the gradient of the graphs show in figure 4 against the factor $\frac{1}{aR}$ has a gradient of $\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)^{-1}$. This graph is shown in Fig 5 and was used to determine the Rydberg constant, as the constant is equal to 1 divided by the product of a times the gradient of the graph.

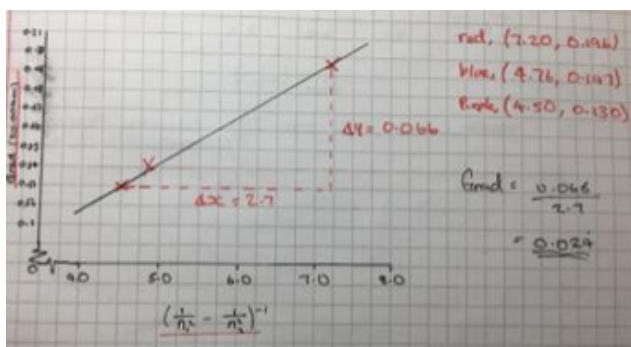


Fig 5. Graph used to determine Rydberg constant

The error on the Rydberg constant was calculated using propagation of error, giving a final value for $R = (1.209 \pm 0.05) \times 10^7 \text{ m}^{-1}$, meaning the result obtained is within three standard deviations of the true value $1.097 \times 10^7 \text{ m}^{-1}$, see ref [1]).

4. Prism and refractive index

The parallel beam of light from the hydrogen lamp will pass symmetrically through the prism if the angle by which the light is refracted is a minimum, called minimum deviation. The angle of deviation, θ , depends on the refractive index of the prism for the wavelength of the light.

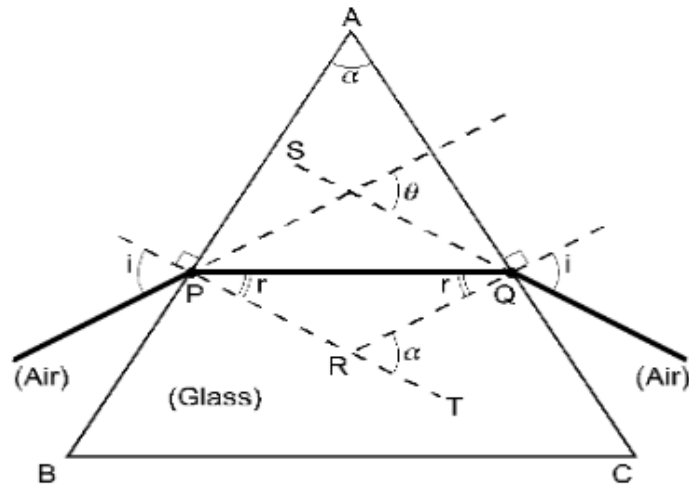


Fig 2. The path of a light ray passing symmetrically through a prism

Since the light is moving through the prism symmetrically the entrance and exit angles are equal. Figure 2 shows that angles AQR and APR are 90 degrees. This means angle TRQ is equal to the prism angle α . By symmetry α is the exterior angle of triangle PQR and is therefore equal to $2r$ where r is the angle of refraction. θ is the exterior angle of triangle SPQ so by symmetry angle PQS and SPQ equal $\theta/2$. Hence the angle of incidence i equals angles SPQ + QPR = $(\alpha+\theta)/2$.

The refractive index is given by snell's law:
$$n = \frac{\sin i}{\sin r} \quad (3)$$

Substituting for i and r gives the equation:
$$n = \frac{\sin\left[\frac{1}{2}(\alpha + \theta)\right]}{\sin\left[\frac{1}{2}\alpha\right]} \quad (4)$$

This allows the refractive index to be obtained by measuring the prism angle α and the angle of deviation θ .

The prism angle was determined by adjusting the prism so that the apex points along the optical axis of the collimator. The light incident on the prism was reflected by the two faces. An image of the slit was visible either side of the slit. The telescope was rotated between these two images and the angle it moved was twice the prism angle α , determined to be 59.879 ± 0.004 degrees.

4.1 Prism setup

The spectrometer was adjusted according to section 3.1. The height of the prism was adjusted so that the parallel light from the collimator enters near the apex. The argument used to determine refractive index of the prism (equation 4) assumes that the ray passes through the prism symmetrically. For this to occur θ must be a minimum. In this experiment the direction of the parallel beam of light cannot be changed. Hence the prism was rotated for each wavelength to change the angle of incidence and ensure the deviation of the beam was a minimum. The angle of minimum deviation was measured for each of the three emission lines by subtracting the angle of the undeviated telescope from the angle at which the parallel beam is refracted by a minimum. These results were used along with the prism angle to calculate the angles of incidence and refraction and hence the refractive index of the prism (using equation 3) for each wavelength of light, red, green and purple.

4.2 Validating the Cauchy relationship

The angle of refraction for each wavelength was half the prism angle (see equation 4) so was found to be 29.940 ± 0.004 . The angles of incidence for red (47.767), green (49.500) and violet (50.583) were found to have an uncertainty of ± 0.004 degrees. These angles were then used with equation 3 to find the refractive index for each wavelength, n^{red} was 1.48, n^{blue} was 1.52 and n^{purple} was 1.54. The uncertainty was added in quadrature to give an overall uncertainty on each refractive index of ± 0.01 .

The Cauchy formula, an empirical relationship giving the dependence of refractive index on wavelength, is $n = A + \frac{B}{\lambda^2}$, where A and B are positive constants, see ref [3].

The refractive index obtained for each wavelength was plotted against λ^{-2} which should be a straight line if the Cauchy relationship is valid. The wavelengths used are shown in Fig 4.

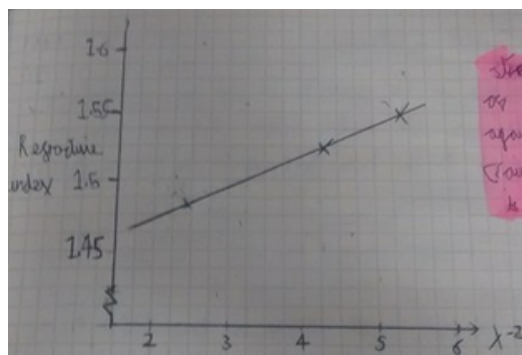


Fig 6. Graph showing validity of Cauchy formula

As shown in Fig 6 the results obtained were consistent with the Cauchy formula as the gradient of the graph is constant.

References

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- [2] Young, H. D. & Freedman, R. A. 2012, University Physics, 13th edition, Addison-Wesley
- [3] Experiments with a Spectrometer lab script, B. Kershaw

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